

Dear Editor,

We would like to submit our manuscript, “Adversarial Vulnerability of Machine Learning GNSS Spoofing Detectors to Transmitted Spoofing Signals,” for consideration as an Original Article in *Satellite Navigation*.

Why this manuscript belongs in *Satellite Navigation*

Machine learning spoofing detectors are moving from research prototypes toward integration in receivers, and the accuracies reported for them are high. Those accuracies are obtained against fixed corpora of recorded attacks, which do not respond to the detector being tested, while a deployed receiver faces an adversary that chooses what to transmit. Our manuscript evaluates fourteen detectors, eight classical and six deep, against attacks that are radio signals selected by an adversary. A counterfeit GPS L1 C/A waveform is synthesised at sample level, summed with authentic recordings from the Texas Spoofing Test Battery and tracked by an open software receiver, so the observables each detector reads are produced by receiver processing of a transmittable signal. The generator is validated by reproducing a recorded attack whose construction is published, and every detector is held at one common operating point with the false alarm rate it pays reported beside every result.

The central finding is that clean detection performance is a poor predictor of resistance to transmitted attacks. The detectors that score best on clean data are among those most often evaded, and the detectors that are never evaded reach that position only by raising false alarms on authentic data to a level a receiver could not accept. We also follow one missed attack through the navigation stage and show that it reaches the timing solution, so detection failure is not confined to the classifier. The manuscript reports the supporting counts, intervals and rates in full. This speaks directly to resilient positioning, navigation and timing, a core scope of the journal, and to the growing literature on learning based GNSS spoofing detection that the journal has published.

Policy issues

We are not aware of any policy issues relevant to this submission.

Competing interests

The authors declare that they have no competing interests.

Prior publication and authorship

This manuscript has not been published and is not under consideration for publication elsewhere, in whole or in part. All authors have seen and approved the manuscript in its current form and agree to its submission to *Satellite Navigation*.

Special issue

This submission is not intended for a special issue.

Thank you for considering our manuscript. We look forward to your response.

Sincerely,

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On behalf of all authors

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